Daniel Taejoon Chong

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Vulnerability researcher and reverse engineer with work for national sponsors including the Air Force Research Laboratory, DARPA, and NHTSA. Builds custom fuzzers, binary emulators, and agentic systems for stripped ARM firmware, and program-analysis tooling in C/C++ and Python. Discovered 160+ bugs in widely deployed software (PHP, CPython, CRuby, NASA CryptoLib), published in peer-reviewed venues, and co-authored a $1.3M SBIR Phase II proposal. Ph.D. candidate (4.0 GPA), completing Spring 2027.

EDUCATION

Ph.D., Computer Science and Engineering (GPA 4.0/4.0) Expected Spring 2027
M.S., Computer Engineering (GPA 4.0/4.0) 2022
B.S., Computer Engineering and **B.S., Computer Science** (GPA 3.93/4.0) 2021
Wright State University, College of Engineering and Computer Science, Dayton, OH

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Researcher (MS/Ph.D.) **May 2021 – Present**
Wright State University, Dept. of Computer Science and Engineering | Advisor: Dr. Junjie Zhang Dayton, OH

Language-Agnostic Taxonomy for Variant Analysis — **23 bugs**
- Built a pipeline that clusters bug reports into an implementation-neutral taxonomy and used it to guide agentic variant hunting across language runtimes.

SeqFuzz: Fuzzing PHP with a Semantics-Aware Intermediate Representation — **68 bugs**
- Designed SeqIR, an IR that captures PHP control and data flow as hierarchical node sequences. Built SeqFuzz to mutate and lift SeqIR into test programs that fuzz the PHP interpreter.

TransFuzz: Cross-Language Fuzzing with LLM-Translated JavaScript PoCs — **50 bugs**
- Built an LLM pipeline that translates historical JavaScript engine PoCs into PHP, with a custom energy scheduler that improves the efficiency of LLM-based mutation.

JotFuzz: Fuzzing the PHP JIT Optimizer — **20 bugs**
- Extracted optimizer-triggering preconditions from historical JIT bugs and generated constraint-satisfying seeds to reach deep optimizer paths.

Function Name Recovery from Stripped x86 Binaries
- Built an AST-like representation lifted from Ghidra P-Code to support function name recovery in stripped binaries.

Reverse Engineer **Oct 2022 – Oct 2023**
Tenet3 Cybernetics Dayton, OH

Hardening of Cyber-Physical Systems
- Reverse engineered ARM, x86, and MIPS binaries to support cyber-physical system supply chain security.
- Developed graph-based binary abstractions for ML-based vulnerability matching across compiled code.
- Built an LLVM-based program mutator that alters binary form while preserving functionality.
- Planned and wrote proposals for project funding rounds; co-authored the $1.3M USAF SBIR Phase II award (AF221-0033).

Zero-Day Bug Hunting and CTF
- Hunted zero-day vulnerabilities and generated exploits to evaluate the effectiveness of company tooling.
- Led the company team in reverse engineering capture-the-flag challenges.

Autonomy Technology Research Center Intern **May 2021 – Jul 2021**
Air Force Research Laboratory (AFRL) Dayton, OH
- Applied symbolic execution and control-flow graph recovery to firmware binaries for security analysis.

Undergraduate Research Assistant **Sep 2019 – Apr 2021**
Wright State University | Sponsor: NHTSA | Project: Vehicle Firmware Security Dayton, OH

- Built a Unicorn-based emulator for stripped ARM32 vehicle firmware, with a peripheral component that feeds simulated device input.
- Created an algorithm that infers the load address of stripped ARM32 binaries (published at SPIoT 2023).
- Combined emulation with symbolic execution into a high-coverage firmware fuzzer.

VULNERABILITIES REPORTED IN OPEN-SOURCE SOFTWARE

PHP (100+), CPython (10), CRuby (8), NASA CryptoLib (2).

HONORS

- WSU 2025 College of Engineering and Computer Science Top Student Award
- WSU 2024 Computer Science and Engineering Graduate Student Excellence Award

GRANTS AND CONTRACTS

Co-author, USAF SBIR Phase II (AF221-0033), "DARWIN Workbench," Tenet3, LLC. $1,299,997 (2024).

PUBLICATIONS

- "TaxoShop: Taxonomy-Guided Variant Analysis Across Programming Languages". To be submitted. **Daniel Chong**, Paschal Amusuo, Junjie Zhang, Zhiqiang Lin.
- "JotFuzz: Fuzzing the PHP JIT Optimizer by Satisfying Optimization Preconditions". Submitted to NDSS 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.
- "TransFuzz: Cross-Language Fuzzing of PHP Interpreters via LLM-Translated JavaScript PoCs." Submitted to USENIX 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.
- "SeqFuzz: Fuzzing the PHP Interpreter with a Semantics-Aware Intermediate Representation." Submitted to FSE 2027. **Daniel Chong**, Junjie Zhang, Zhiqiang Lin.
- "FedPDFGuard: A Federated Learning Approach for Robust Detection of Malicious PDF Files." Proceedings of the IEEE National Aerospace and Electronics Conference. Pengyu Zhou, **Daniel Chong**, Honglu Jiang, Junjie Zhang, Rui Dai. July 2025.
- "An Empirical Study of Software Sanitization Locality." Proceedings of the IEEE National Aerospace and Electronics Conference. Nathaniel Boland, Junjie Zhang, **Daniel Chong**, and Rui Dai. July 2024.
- "Automatically Inferring Image Base Addresses of ARM32 Binaries Using Architecture Features." Proceedings of the 12th International Symposium on Security and Privacy On Internet Of Things. **Daniel Chong**, Junjie Zhang, Nathaniel Boland, and Lingwei Chen. November 2023.

PRESENTATIONS, TEACHING, AND SERVICE

- Poster: "TaxoShop: Taxonomy-Guided Variant Analysis," KOCSEA, 2026
- Talk: "Automatically Inferring Image Base Addresses of ARM32 Binaries Using Architecture Features," SPIoT, 2023.
- Talk: "Reverse Engineering ARM32," Great Lakes Region Aerospace Education Conference, 2023.
- Graduate Teaching Assistant: Trust Networks (Fall 2023, Spring 2024); Introduction to Quantum Computing (Fall 2022), including supplemental research discussions.
- The Network and Distributed System Security Symposium 2027 (External Reviewer)

TECHNICAL SKILLS

- **Custom fuzzing harnesses:** designed and built three fuzzers (IR-guided, constraint-guided, and LLM-driven) for language interpreters and a JIT optimizer, plus a high-coverage ARM firmware fuzzer.
- **Binary emulation and dynamic analysis:** Unicorn-based ARM32 emulator with peripheral emulation; symbolic execution and taint analysis of firmware.
- **Disassemblers and decompilers:** Ghidra, IDA, Radare2, Capstone.
- **Embedded RE:** ARM32, ARM64, MIPS, and x86-64 binaries from cyber-physical and vehicle systems; image-base inference for stripped ARM32 firmware.
- **Languages and tooling:** C/C++, Python, PHP, JavaScript; ARM, MIPS, and x86 assembly; LLVM (IR design, program mutation); AFL/AFL++; symbolic execution and taint analysis; LLM-guided program transformation.
- **Proposal and technical writing:** co-authored a $1.3M USAF SBIR Phase II proposal; 7 papers (3 published, 4 under review).